



Carisurg

Assignment 11

Interim Proposal

Due Date: June 20th, 2026.

Prepared by **Josiah-John Green**

Project Context

This project is structured as two phases, referenced repeatedly throughout the proposal, diagrams, and weekly write-ups. They are not interchangeable — each has a different scope, timeline, and deliverable.

Phase 1 — the 12-week pilot (this programme). The goal is to design and validate a structured **digital triage capture tool** that mirrors the existing Mercer General paper triage form. The output is not an AI model — it is a clean, de-identified, queryable dataset built from real ED encounters. Mercer currently runs entirely on paper, which means no structured triage data exists anywhere to train or validate a model against. Phase 1 exists to fix that gap first.

Phase 2 — beyond this programme (future work). Once a sufficient digital dataset exists from Phase 1, Phase 2 would use that data to train and locally validate a **rule-based ESI classifier**, iterating toward a machine learning model as data volume grows over time. This is the long-term vision the project is building toward — but it is explicitly *not* a deliverable of the current 12-week pilot.

Why the split matters: Most AI-assisted triage literature assumes infrastructure — electronic health records, stable data pipelines — that Mercer does not have. Proposing an AI classifier without first proposing the data foundation it depends on would be building on top of nothing. Phase 1 is the foundation. Phase 2 is the building.

Workflow Notes

The Mercer General Emergency Department follows a door-to-disposition pathway with five measured timestamps: **arrival**, **triage**, **seen-by-physician**, **disposition decision**, and **exit**.

Stage	Duration	What Happens	Data Captured
Arrival	—	The patient arrives via ambulance, walk-in, or private vehicle	None yet
Registration	2–3 min	The patient’s data, including their demographics, consent, and arrival modality, is collected.	Front desk — paper intake form
Triage Assessment	3–5 min	The patient’s severity is assessed based on chief complaint, vitals, allergies, pain score, and ESI level assigned.	Triage nurse — paper front page
Zone Placement	—	The patient is placed in the ESI zone. Zones 1–3 equate to bed placement (resus/acute); while	None — disposition only

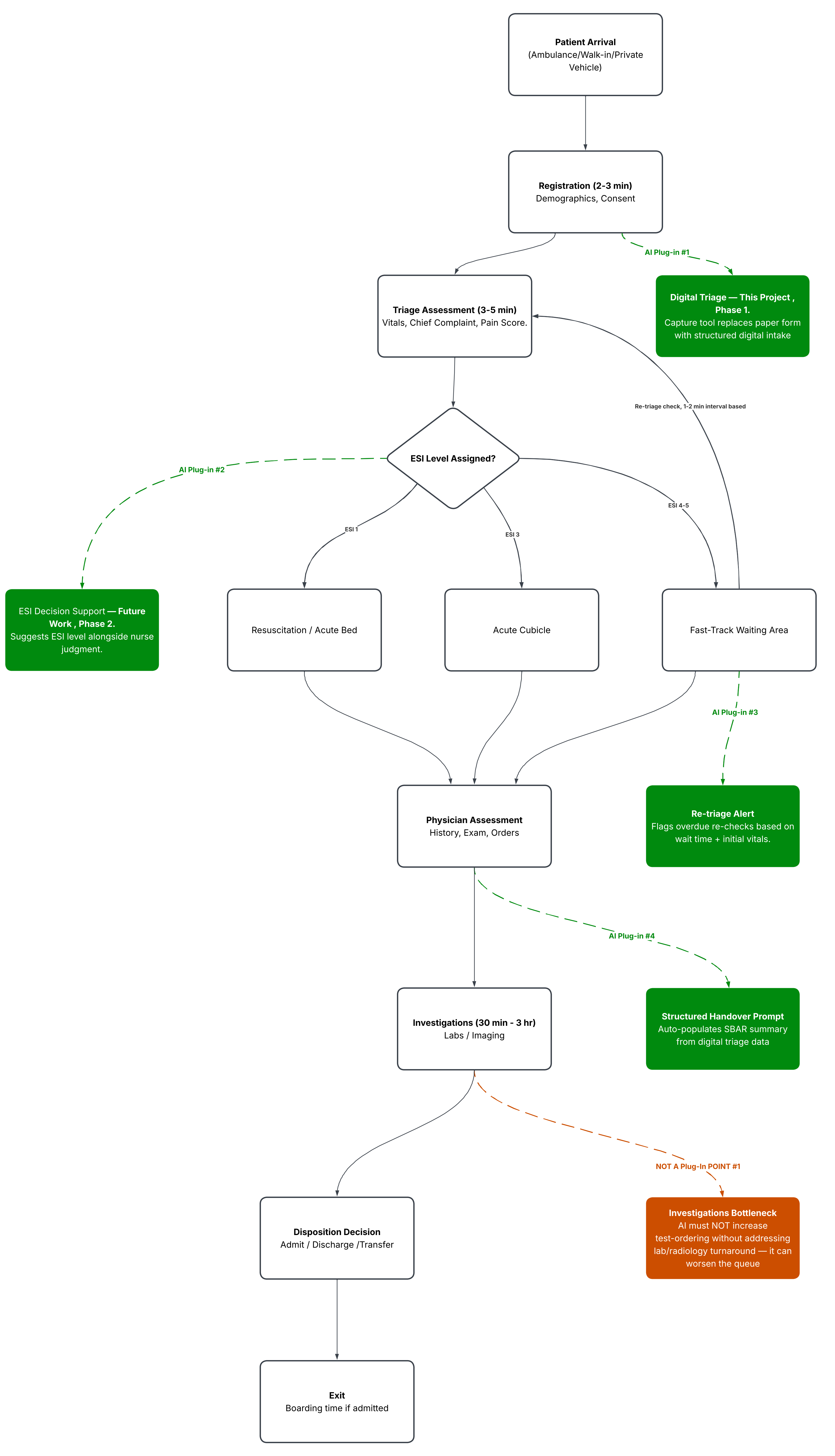
		zones 4–5 equate to fast-track waiting.	
Physician Assessment	ESI 2: ≤10 min ESI 3: 30–60 min	The physician accesses patient, collates their medical history, does examinations, and places orders for same.	Physician — deeper clinical pages
Investigations	30 min – 3 hr	The labs/imaging are processed, collected and accessed — shared hospital resource.	Lab/radiology system
Disposition Decision	At the end of the workup	The patient is either admitted/discharged/transferred based on the recommendation by the physician.	Physician
Exit	Variable	The patient leaves the hospital; however, this is delayed if admitted.	—

Key observations on information loss:

1. The triage front page (ESI, vitals, chief complaint) is **consistently completed**.
2. The deeper clinical pages (history, medication chart, disposition rationale) are **not consistently completed** — this is the first place structured data goes missing.
3. **Re-triage** depends entirely on a nurse visually scanning the waiting room at intervals. There is no structured trigger for re-assessment — a deteriorating ESI-4 patient is only caught if someone happens to notice.
4. Investigations are the dominant flow bottleneck. Lab and radiology are shared with the rest of the hospital, and ED workups compete directly with elective inpatient demand.

Workflow Diagram

In the diagram below, decision diamonds are placed at the triage-level ESI assignment step, per task instructions. Along with five annotated AI plug-in points, including one explicit **non-plug-in point** — a place where adding AI would actively make things worse.



Workflow Constraints

Constraint 1 — The 3–5 Minute Triage Window (Friction Budget)

Triage nurses have a narrow assessment window and a queue of waiting patients behind every encounter. Any digital tool that asks for more fields than the existing paper form, or takes more time than handwriting, will be abandoned within days — regardless of how good the underlying model is. This directly shapes the interface design for the Phase 1 capture tool: it must replicate the paper form's speed, not add to it.

Constraint 2 — Paper-to-Digital Lag at the Point of Capture

All triage data currently exists only on paper, and even there, only the front page (ESI, vitals) is reliably completed. Before any classifier — rule-based or ML — can be trained or validated, structured digital data must exist at the moment of capture, not as a later transcription step prone to loss and delay. This is the foundational justification for the entire Phase 1 proposal.

Constraint 3 — Shared Investigations Bottleneck

Labs and radiology are shared hospital-wide resources with 30-minute to 3-hour turnaround, and ED workups compete directly with elective inpatient demand. This is the dominant flow bottleneck at Mercer. Any AI design that increases test-ordering volume without addressing turnaround capacity will worsen overcrowding rather than relieve it — this constraint actively rules out certain plug-in points (see Investigations stage in diagram above) rather than just shaping their design.

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal.

Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)

MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring the level of consciousness
RR	Respiratory Rate — breaths per minute
FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
CTAS	Canadian Triage and Acuity Scale — a triage system similar to ESI, used as a comparison benchmark in some studies
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

LOS	Length of Stay — total time a patient spends in the ED, from arrival to exit
DNR / DNI	Do Not Resuscitate / Do Not Intubate — patient code status designations

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations (e.g. Vanuatu, many Caribbean states) facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. "admit") and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 (no better than chance) to 1.0 (perfect); higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm that builds an ensemble of decision trees, commonly used for structured/tabular data like vitals
Rule-based classifier	A decision-making system using explicit if/then logic (like the Day 6 triage pseudocode) rather than a trained statistical model

Multimodal	Combining more than one data type (e.g. structured vitals + unstructured free-text notes) in a single model
-------------------	---

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers (e.g. https://doi.org/...)
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A type of literature review that maps the breadth of existing research on a topic, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol is often used to assess evidence quality

Preprint	A paper shared publicly before completing formal peer review (e.g. via ResearchSquare)
-----------------	--

Version Control & Engineering Terms

Term	Meaning
PR	Pull Request — a proposed set of code changes submitted for review before merging into the main branch
Feature branch	A separate copy of a codebase is used to make changes safely without affecting the main version
Mermaid	A text-based diagramming syntax that renders automatically as a flowchart/diagram when viewed on GitHub
.gitignore	A file listing what should NOT be tracked by version control (e.g. patient data, local environments)